Time to Power:
The Financing Cost of Energy-Delayed Megaprojects

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Abstract

We assemble an audited dataset on 19 large capital-intensive projects (19 rows, 6 sectors: data centers, semiconductors, mining, LNG, battery manufacturing, industrial processing), nine of which were materially affected by energy-infrastructure constraints, and ask when the cost of carrying financing obligations on a delayed asset exceeds the cost of directly funding the missing power. A 12-mechanism delay-cost model, applied to a stylized 500 MW / $6.0B data center, puts a 12-month energy-driven delay at $857.6M in present value (14.3% of capex, $71.5M per month); the timing value of delayed revenue is the largest term at every horizon (about two-fifths of the total), while financing-only terms are 35% of the 12-month total, so counting only interest during delay understates cost by roughly two-thirds. Five independent energy supply-chain links each run 30 months or more in typical lead time (greenfield transmission 120 months, generator interconnection queues 55 months median, gas-turbine slots 5–7 years quoted, large transformers 30 months average with 48–60 month Tier-1 quotes, large-load interconnection 36 months typical). A wait-vs-solve model over 3-, 6-, 12-, 24-, and 36-month delays finds that no permanent alternative beats waiting at 12 months or less; reciprocating gas engines are the only alternative that beats waiting inside 60 months, from roughly 24–28 months of expected delay, supporting a rational capex premium up to $3,049/kW at 36 months (against $1,600/kW market capex) and a delivered-energy ceiling near $78/MWh. The data support a durable but narrow business thesis: willingness to pay for time-to-power is large (order $140/kW per month of acceleration) yet confined to delays beyond roughly two years, alternatives priced near grid power, and revenue-intense projects. We state the dashboard readings that would prove this wrong. All figures are nominal USD, base year 2026, unless stated.

JEL: D24, L94, Q41, G32.

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1 Introduction: the research question

A growing class of capital-intensive projects reaches financial close with tens of billions of dollars committed and then cannot start: the electrons, molecules, wires, or permits that would let the asset operate are not there. The committed capital still carries its financing cost. Interest accrues, commitment fees meter, revenue that was contracted to begin on energization does not, and the equity sponsor watches its internal rate of return decay month by month.

The question this paper answers is narrow and quantitative: under what circumstances does the cost of continuing to carry financing obligations on the delayed asset exceed the cost of directly financing, building, acquiring, or contracting the missing energy infrastructure? Put differently, at what delay horizon should a rational owner stop waiting for the grid and buy the power itself, and how much should it rationally pay for speed?

Three prior observations motivate the question. First, the financing is real and large: the 19 projects in our database carry committed stacks up to $18.4B at a single final investment decision (Rio Grande LNG, NextDecade company release (NextDecade Corporation 2023), Observed, grade A). Second, the energy delays are real and long: nine of the 19 projects were materially affected by power or energy constraints, with market-level moratoria running to 48 months. Third, owners are already answering the question with their own balance sheets: xAI bridged a 150 MW grid shortfall at Colossus with about 35 mobile gas generators rather than wait, and Crusoe ordered 29 aeroderivative turbines for Abilene (Company and press reporting 2026). The market is pricing time-to-power before the literature has measured it.

We do not ask whether any particular security is mispriced. We ask what the data say about the cost of waiting, the cost of solving, and the boundary between them. This is a retrospective study: every estimate describes the historical and audited sample; the only forward-looking statements are the explicitly labelled break-even horizons, which are conditional on the stylized project's parameters, not forecasts.

The remainder of the paper is organized as follows. Section 2 reviews the related literature; Section 3 defines the data and the unified evidence taxonomy; Section 4 states the delay-cost and wait-vs-solve models and the stylized project. Section 5 reports the project database, the bottleneck lead times, the twelve cost mechanisms, the alternative supply options, the break-even boundary, the real cases, and the revenue and fuel-price sensitivity. Section 6 determines the thesis: who pays, how narrow the stranded-capital pool is, the bottleneck hierarchy, willingness to pay, the verdict, and the falsifiers. Section 7 states the robustness limits; Section 8 concludes; Section 9 documents the dataset. References follow.

2 Related literature

No prior study measures in full the financing cost of an energy-driven delay on a capital-intensive project, or the rational capex premium for buying time-to-power. The literature bearing on the question sits in four strands, and the gap between them is this paper's subject.

First, construction-cost and delay benchmarks. Turner & Townsend's international construction cost survey (Turner & Townsend 2025) and the EIA's overnight-cost baselines (U.S. Energy Information Administration 2020, 2025) document what projects cost to build and how those costs escalate; they do not price the carrying cost of the committed capital while the asset sits idle. Lazard's levelized cost of energy (Lazard 2025) benchmarks what generation costs to own, not what the delay costs the load it was built to serve.

Second, the interconnection and permitting record. FERC's large-load interconnection proceedings (Federal Energy Regulatory Commission 2025a) and NERC's reliability assessments (North American Electric Reliability Corporation 2025) describe the queue and capacity constraints that generate the delays we cost; the Atlantic Council's (Atlantic Council 2026) permitting work documents the transmission timelines behind them. These sources measure the

wait; none converts the wait into the owner's carrying cost.

Third, the data-center market's own pricing of speed. CBRE (CBRE 2026) and Cushman & Wakefield (Cushman & Wakefield 2026) report the lease rates, vacancy, and build costs from which the market's willingness to pay for delivered power can be read directly; the press record (Company and press reporting 2026) shows owners bridging with mobile generation and turbine orders rather than waiting. These are the revealed preferences our model rationalizes.

Fourth, the corporate-finance treatment of idle capital. Interest during construction is capitalized in project accounts and commitment fees are disclosed in credit facilities, but the full twelve-mechanism cost of delay, including foregone revenue timing value, hedge costs, and contracted-power exposure, is not a standard reporting line. This paper's contribution is to price the wait and the solve on one balance sheet, with falsifiable thresholds.

3 Data and sources

3.1 Source-quality grading and the data taxonomy

Every figure in this paper carries a two-part evidence tag: a data status and a source-quality grade. Data status is **Observed** (a value taken directly from a source as published, without transformation), **Reconstructed** (values assembled from primary fragments or re-tabulated by stated methods), **Modelled** (values computed from stated inputs, including the delay-cost and wait-vs-solve engine outputs), or **Assumed** (parameters fixed by the analyst with stated reasoning, where no observation exists). Source quality is **A** (official primary: company releases, regulatory filings, government statistics), **B** (reputable secondary: established press, analyst research, guided company figures), or **C** (provisional: estimates, single observations, or model outputs presented as planning inputs). Tags are never mixed silently: a Modelled value built on an Assumed input inherits the provisional grade. Numbers that could not be verified were withheld rather than printed; blank cells in the datasets are deliberate.

3.2 The four datasets and two models

master_projects.csv (19 rows, 30 columns) records real projects with large committed capital stacks across data centers (7), semiconductors (3), mining (2), LNG (3), battery manufacturing (3), and industrial processing (1). For each project we reconstruct, where verifiable: project debt, corporate debt, sponsor equity, lease obligations, construction loans, interest expense, commitment fees, capitalized interest, required returns, completion guarantees, contractual delivery dates, revenue commencement dates, penalties, and force-majeure provisions. Nine rows carry an energy-delay flag across four sectors; one further row (xAI Colossus) is flagged partial, the counterfactual case in which the constraint was real but the schedule held because the tenant built the power itself.

master_energy_bottlenecks.csv (18 rows) covers 11 supply-chain links: new-build gas generation, grid allocation, fuel supply, gas pipeline laterals, new interstate pipelines, greenfield transmission, reconductoring, substations, large power transformers, generator interconnection queues, large-load interconnection, battery storage, backup gensets, water supply, chillers, and permitting (NEPA, air, and state/local split out). Each row records typical lead time, range, historical delay evidence, an estimated delay probability, points of failure, and workaround options. Every delay probability in this dataset is Assumed, grade C: no primary source publishes schedule-miss probabilities, and the paper presents them as planning assumptions with stated reasoning, not measured statistics.

master_delay_mechanics.csv (12 rows) formalizes the twelve mechanisms by which delay becomes cost, each with a closed-form formula implemented in *results/delay_cost_model.py*.

master_alternatives.csv (24 rows) compares the alternatives a project owner could fund directly: reciprocating gas engines, aeroderivatives, combined-cycle gas, gas pipeline laterals,

Table 1: Dataset and model inventory: summary statistics.

Dataset / model	Rows	Coverage	Headline
`master_projects.csv`	19	6 sectors; 9 energy-delayed + 1 partial	$18.4B max stack at F
`master_energy_bottlenecks.csv`	18	11 supply-chain links	30–120 mo lead times
`master_delay_mechanics.csv`	12	12 cost mechanisms	$W(12)$ = $857.6M
`master_alternatives.csv`	24	17 alternative supply options	recip engines $58/MW
`delay_cost_model.py`	—	T = 3–36 mo	12-mechanism $W(T)$
`wait_vs_solve.py`	—	per-alternative break-evens	$3,049/kW max 36-mo

Notes: Evidence tags per Section 3; headline outputs in `results/key_estimates.csv` (103 estimates).
Source: Author's compilation; four master CSV files and two Python models in `data/` and `results/`.

solid oxide fuel cells, the full hydrogen chain, batteries, solar, solar-plus-storage, wind, wind-plus-storage, large nuclear, small modular reactors, conventional and enhanced geothermal, microgrids, rental gensets, and customer-paid grid upgrades. Each row is evaluated on delivered cost per MWh, capital intensity, construction lead time, reliability, permitting burden, fuel security, scalability, and demonstrated ability to serve hundred-megawatt to gigawatt loads.

The two models: *results/delay_cost_model.py* computes the present-value cost of a delay of T months decomposed by mechanism, and *results/wait_vs_solve.py* compares that waiting cost against the cost of funding each alternative directly, over 3-, 6-, 12-, 24-, and 36-month delay scenarios. Headline model outputs are frozen in *results/key_estimates.csv* (103 estimates), the machine-readable contract the audit checks. Table 1 summarizes the datasets and models.

3.3 Identification: two projects, not one

The commissioning brief named "Project Jupiter / Oracle / Blue Owl" as a case study. The evidence does not support treating that phrase as one project. Two separate developments are involved, and the database keeps them as separate rows:

Abilene, Texas (PRJ-DC-02): the Crusoe/Blue Owl development on the Lancium Clean Campus, 1.2 GW across eight buildings, about $15B of debt and equity (about $9.6B in JP-Morgan loans plus about $5B of Crusoe/Blue Owl cash), Oracle as 15-year leaseholder, OpenAI as compute customer. The first two buildings were operational in September 2025; power is grid plus on-site gas turbines. No verified energy-driven delay. Tier: Observed, grade B.

Project Jupiter, Doña Ana County, New Mexico (PRJ-DC-01): the Stack Infrastructure (a Blue Owl unit) development, 2.45 GW campus, about $18B in syndicated loans from roughly 20 banks, quoted at 89–91 cents on the dollar in secondary trading, Oracle as anchor tenant, target online 2028. The recorded delay is a roughly six-month slip in the natural-gas pipeline in-service date (August 2026 to February 2027) after the New Mexico State Land Office denied right-of-way twice; a separate air-quality permit for the Bloom Energy fuel-cell system remained pending; Oracle invoked force majeure on 24 September 2026, preserving the right to defer payments if the 2028 target is missed. Tier: Observed, grade B. The campus target itself has not officially slipped; any Modelled cost below is conditional on the pipeline slip becoming an energization slip.

Nothing in this paper depends on either row alone. Project Jupiter is one of nine energy-delayed rows and one of 19 rows total.

4 Methodology

4.1 The delay-cost engine

For a delay of T months beginning at the planned revenue-commencement date, the incremental present-value cost is

$$W(T) = \sum_{i \in \{1..8,11,12\}} m_i(T), \tag{1}$$

where the m_i are the twelve mechanisms of Table 2, discounted at a single project after-tax WACC. Mechanisms 9 (sponsor IRR erosion) and 10 (equity dilution) are memo metrics, not additive terms: they re-express costs already counted, from the sponsor's perspective. Interest accrual is split between cash interest (m_1) and capitalized interest (m_3) by the capitalization share so it is never counted twice. Delayed rent (m_7) is computed on EBITDA, not on net-of-debt cash flow, because debt service in the window is already in m_1 and m_3. Opportunity cost (m_{12}) covers equity only. All present values use one numeraire: the project WACC, 6.52% for the stylized case. Tax shields are excluded on both sides consistently.

Table 2: The twelve delay-cost mechanisms. T in months.

ID	Mechanism (formula)
m1	Cash interest on drawn debt: $D_{\text{drawn}} \times r \times T/12 \times (1-\kappa)$
m2	Commitment fees on undrawn capital: $D_{\text{undrawn}} \times f \times T/12$
m3	Capitalized construction interest: $D_{\text{drawn}} \times r \times T/12 \times \kappa$
m4	Stressed refinancing: PV of margin shock over tenor plus fees, only if maturity falls in the window
m5	Loan/bond repricing on delay news: PV of step-up over tenor, only past the trigger
m6	Covenant waiver and legal cost, only past the grace period
m7	Delayed rent: PV of the EBITDA timing shift
m8	Permanently lost revenue, discounted
m9	Sponsor IRR erosion (memo metric)
m10	Equity dilution (memo metric)
m11	Construction-cost inflation on unspent capex
m12	Opportunity cost of committed equity

Notes: m9 and m10 are memo metrics (IRR erosion and dilution), not additive cost terms.
Source: Author's model; formulas in `results/delay_cost_model.py`.

4.2 The wait-vs-solve model

For an alternative with lead time L months, the solve cost is

$$S(D, \text{alt}) = W(L) + K_{\text{alt}} + E_{\text{alt}}, \tag{2}$$

where $W(L)$ is the waiting cost that still accrues while the alternative is built, K_{alt} is alternative capex scaled to the MW shortfall (assumed spent at mid-construction), and E_{alt} is the present value of $(\text{delivered}_{\text{alt}} - p_{\text{grid}}) \times$ annual MWh over a 20-year operating life, starting at energization. The grid comparator price is \$60/MWh (Assumed, grade C). The alternative is modeled as a permanent behind-the-meter asset kept for the full operating life; retiring it when grid power arrives would lower S and is not modeled, which is conservative against the solve case.

The break-even output is the maximum rational capex premium for time-to-power,

$$\text{max_capex}(D, \text{alt}) = W(D) - W(L) - E_{\text{alt}}, \tag{3}$$

the marginal cost of the months saved, net of the lifetime energy premium, stated per kW of accelerated capacity and as a maximum sustainable $/MWh premium over grid. The break-even horizon D^* is found by bisection on $W(D) - S(D, \text{alt})$.

4.3 The stylized project

All modeled figures use one stylized project: a 500 MW data center, $6.0B capex ($12.0M/MW, inside the $10.7M/MW Turner & Townsend (Turner & Townsend 2025) and $8.9–23.3M/MW Cushman & Wakefield (Cushman & Wakefield 2026) ranges), 70% debt at 6.37% (SOFR 3.87%, Observed via FRED (Federal Reserve Economic Data (FRED) 2026), plus a 250 bp margin, Assumed), 37.5 bp commitment fee on undrawn capital, $1.0B annual revenue ($2.0M/MW/yr, the low end of CBRE asking rents (CBRE 2026) at $1.98–2.46M/MW/yr, discounted for lease-up), 50% EBITDA margin, 5.0% construction inflation (BLS PPI, 5.1% in the twelve months to August 2026 (U.S. Bureau of Labor Statistics 2026)), 60% of capex spent at delay start, 10% equity required return, 6.52% after-tax WACC. These are stated parameters, not observations; the model is linear enough in most of them that the sensitivities in Section 5.7 show where the conclusions move.

5 Results

5.1 The project database: capital stacks and energy delays

The 19 database rows span the full capital-stack shapes the research question requires. The strongest reconstructions:

Rio Grande LNG Phase 1 (PRJ-LNG-03): $18.4B at final investment decision on 12 July 2023, every tranche disclosed: $11.6B senior secured bank facilities ($11.1B construction term loans plus $0.5B working capital), $700M private placement notes, about $5.9B of JV equity (GIP, GIC, Mubadala, TotalEnergies), $283M NextDecade equity, Bechtel EPC of about $12B, and about $3.1B of interest during construction. Tier: Observed, grade A (NextDecade company release (NextDecade Corporation 2023)).

Stargate Abilene (PRJ-DC-02): about $15B across two JPMorgan loans (one $7.1B tranche) plus sponsor cash; a second-phase raise of $11.6B managed by Blue Owl Real Assets and Primary Digital Infrastructure. Tier: Observed, grade B.

xAI Colossus (PRJ-DC-03): about $18B of GPU spend and a $20B Series E raise reported. Tier: Observed, grade B.

Northvolt Ett (PRJ-BAT-02): about $15B raised across debt, equity, and grants; $5.8B of debt and $30M of cash at the Chapter 11 filing. Tier: Observed, grade B.

Nine rows are flagged as materially affected by energy constraints, across data centers, battery manufacturing, industrial processing, and mining:

Data centers. Project Jupiter, New Mexico: the six-month pipeline slip and pending fuel-cell air permit described above, with the Oracle force-majeure notice of 24 September 2026. Dublin: EirGrid halted new data-center grid connections in 2022, a de facto moratorium to 2028. Northern Virginia: Dominion's transmission pinch in eastern Loudoun, disclosed July 2022, with some projects delayed as many as four years. Singapore: government moratorium on new builds from 2019 to January 2022 (about 36 months). Amsterdam/Haarlemmermeer: municipal halt from July 2019, lifted after about one year.

Battery manufacturing. Agratas Somerset, UK: a £4B plant (about $5.3B at the reporting outlet's own conversion) whose National Grid substation arrives in late 2028, about 12 months after the phase-1 production target of late 2027; two temporary grid connections bridge the interim. The causal link between the substation timing and the production-slip timeline is not established in the sources found; both facts are recorded separately. Tier: Observed, grade B.

Industrial processing. Alcoa San Ciprián, Spain: curtailed in 2021 on power prices; the Alcoa/Ignis restart delayed to mid-2026 after the 28 April 2025 Spain-wide blackout disrupted the smelter and alumina refinery. Alcoa guides a $90–110M net loss in 2025 and $110–130M of operating cash used. Tier: Observed, grade B (company disclosures (Alcoa Corporation 2025)).

Mining. First Quantum, Zambia: ZESCO curtailed 80 MW from May to December 2024 during drought; force-majeure notice 11 April 2024; $25M of additional power-sourcing cost estimated for the remainder of 2024. Tier: Observed, grade A (company release and MD&A (First Quantum Minerals 2024)). Kamoa-Kakula, DRC: grid instability cut 2024 guidance from 440–490kt to 425–450kt; 36,500t of copper production lost; 220 MW of diesel backup installed. Tier: Observed, grade B. The mining rows are operating assets with curtailed output, not delayed greenfield starts; their delay-month cells are blank and the flag records materially affected production schedules and cost.

The partial case, *xAI Colossus* (Memphis): the local grid could not supply 150 MW on the tenant timeline, so xAI bridged with about 35 mobile gas generators (up to 420 MW), Tesla Megapacks, and a 150 MW substation, and plans a permanent 1.2 GW gas plant by July 2027. The constraint was real and material; the schedule held because the tenant built the power itself. It is the counterfactual the model needs: the cost of waiting against the cost of self-supply, observed in the field.

Comparator rows without energy delay (Intel Ohio One, TSMC Arizona, Samsung Taylor, LNG Canada, Mozambique LNG, Northvolt, VW PowerCo St. Thomas) give the paper non-energy delay causes, so the energy cases are priced against something.

5.2 Bottlenecks: how long the links take

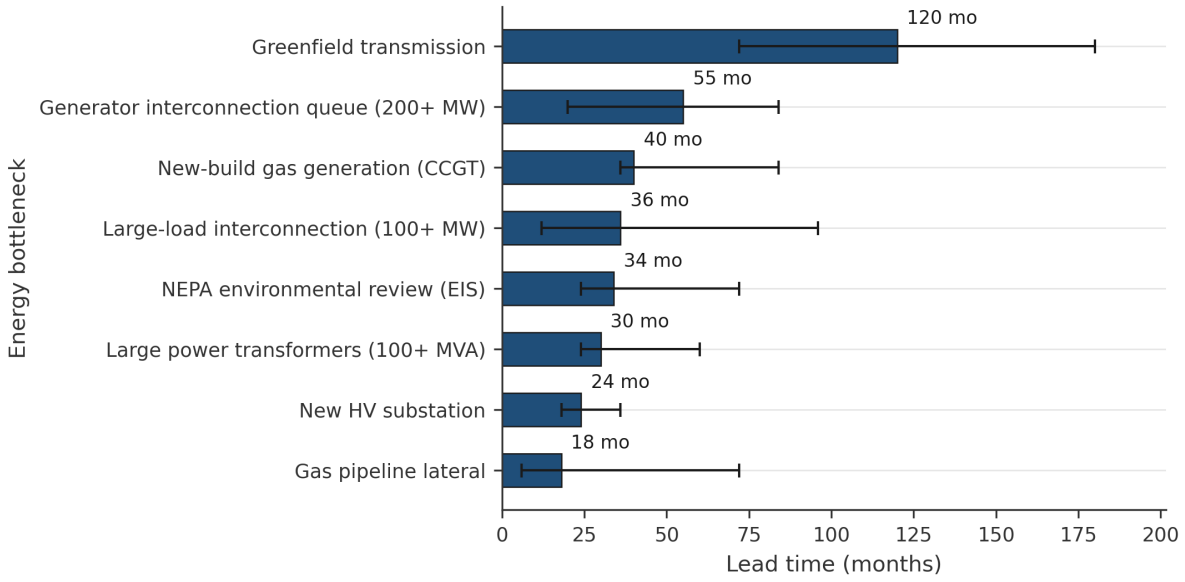


Figure 1: Typical lead times for energy supply-chain links, with ranges. Five independent links run 30 months or more typical; three are at or above 55 months. Delay probabilities are Assumed, grade C, and are not plotted as measurements.

Figure 1 shows the lead-time evidence. Five independent links each run 30 months or more in typical lead time; three are at or above 55 months:

New greenfield transmission: 120 months typical, range 72–180 (NERC 6–15 years (North American Electric Reliability Corporation 2025), Observed, grade A). The Atlantic Council measured an average 6.5-year permit wait before construction begins (Atlantic Council 2026);

SunZia took 17 years to win approval. Only 322 miles of 345 kV+ lines were completed in the US in 2024, the third-slowest year in 15.

Generator interconnection queues (200+ MW): 55 months median from request to commercial operation (LBNL Queued Up 2024 (Lawrence Berkeley National Laboratory 2024), Observed, grade A); only 19% of 2000–2018 entrants reached operation by end 2023, and more than 70% withdrew. The 2025 edition puts the median above 4 years across 2,061 GW in queue. Network upgrade costs averaged 70% of total interconnection costs for withdrawn projects. PJM projects entering service in 2025 averaged more than 7 years to energize.

New-build gas generation: 40 months in the EIA reference case for a 1×1×1 H-class CCGT (U.S. Energy Information Administration 2025) (Observed, grade A), but heavy-frame turbines are now quoted at 5–7 years from order to commercial operation, with GE Vernova, Siemens Energy, and Mitsubishi Power order books sold out years ahead (Observed, grade B (Company and press reporting 2026)).

NEPA environmental review: 34 months median from notice of intent to final EIS, 2019–2024, n=1,903 (Council on Environmental Quality, January 2025 (Council on Environmental Quality 2025), Observed, grade A).

Large power transformers (100+ MVA): about 30 months average in mid-2025 (Wood Mackenzie (Wood Mackenzie 2025), Observed, grade B), up from about 50 weeks in 2021, with prices up 77% since 2019. Tier-1 OEMs (Hitachi Energy, Siemens Energy, GE Vernova) now quote 48–60 months with backlogs through 2029–2031; Wood Mackenzie projects a 30% US supply deficit in 2026 easing to about 5% by 2030.

Large-load interconnection (100+ MW demand): 36 months typical, range 12–96 (Observed, grade B). The US median grid-connection wait reached 5 years in 2025 against about 18 months in 2015; the UK demand-connection queue grew 460% in the first half of 2025 with waits to 15 years. FERC's October 2025 notice of proposed rulemaking targets 90-day processing of large-load requests (Federal Energy Regulatory Commission 2025a), but the downstream build (substations, transformers, breakers) is unaffected by study reform.

A megaproject that needs more than one of these links in series faces a stacked schedule no financing structure can shorten. Delay probabilities (60–80% for queues against sub-4-year targets, 30–50% for transformers, 50–70% for greenfield transmission, 40–60% for large-load interconnection) are Assumed inputs with stated reasoning in the dataset, grade C throughout.

5.3 Financial transmission: the twelve mechanisms at work

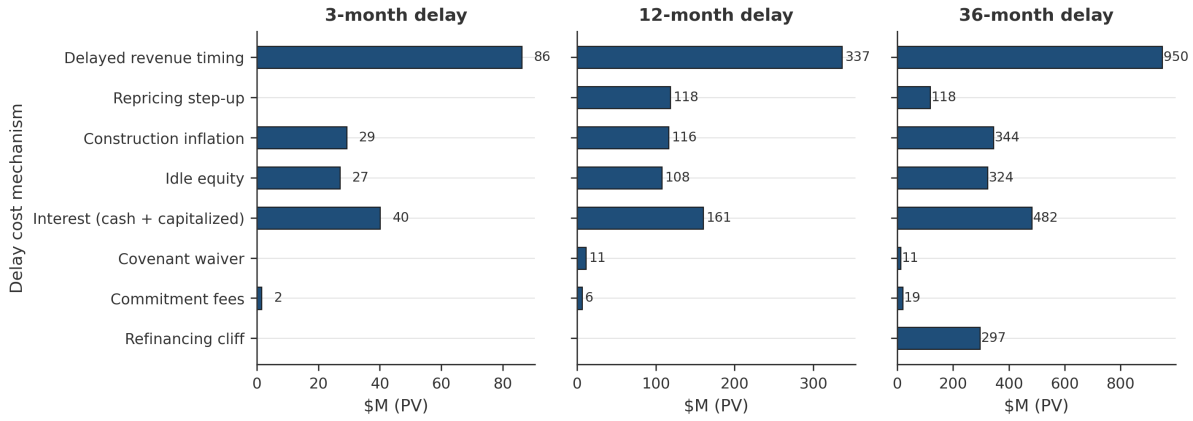


Figure 2: Incremental present-value cost of delay for the stylized 500 MW / $6.0B data center, decomposed by mechanism, at 3, 12, and 36 months. Mechanism 7 (delayed revenue timing) is the largest term at every horizon. Financing-only terms (m1–m6) are a minority of the total.

Table 3 and Figure 2 give the stylized results. A 12-month delay costs $857.6M in present value: 14.3% of capex, $71.5M per month. The decomposition:

Table 3: Waiting cost $W(T)$ for the stylized 500 MW / $6.0B data center.

Delay	$W(T)$	Per month	Share of capex	Sponsor IRR
3 mo	$184.1M	$61.4M	3.1%	10.37% → 9.81%
6 mo	$485.1M	$80.8M	8.1%	—
12 mo	$857.6M	$71.5M	14.3%	10.37% → 7.33%
24 mo	$1.86B	$77.5M	31.0%	—
36 mo	$2.54B	$70.7M	42.4%	10.37% → 3.85%

Notes: PV at 6.52% WACC. All Modelled, grade C.
Source: Author's model, `results/delay_cost_model.py`; parameters in Section 4.3.

At 12 months the largest terms are delayed revenue timing (m7, $336.7M), loan repricing (m5, $118.3M), construction inflation (m11, $116.3M), idle-equity opportunity cost (m12, $108.0M), and interest (m1 plus m3, $160.6M combined). Covenant waiver and commitment fees are small; refinancing and lost revenue do not fire in the base case. Sponsor equity IRR falls 3.05 points to 7.33%; delay cash costs of $322.3M exceed the $300M contingency by $22.3M, forcing fresh equity and 1.2% dilution.

Three regularities survive the parameter choices. First, the timing value of delayed revenue (m7) is the largest single term at every horizon, roughly two-fifths of the total. Second, per-month cost is nearly flat ($61–81M) rather than rising, because fixed event costs (m5 repricing) front-load short delays while the maturity cliff (m4, $296.7M at 24 months) is binary. Third, the financing-only terms (m1–m6) are 35% of the 12-month total; the rest is revenue timing, inflation, and idle equity. Any wait-vs-solve comparison that counts only interest during delay understates the cost by roughly two-thirds.

5.4 Alternatives: what the owner could build instead

Figure 3 compares the alternatives. Technology cost and performance parameters follow the EIA technology baselines (U.S. Energy Information Administration 2020, 2025) except where a primary source (e.g., the Net Power 8-K (Net Power Inc. 2026)) supersedes them. For a 500 MW need on a three-year horizon, the binding constraint is schedule, and the options that can plausibly reach full output inside 36 months are:

Reciprocating gas engines (Wärtsilä 50SG class): 14–18 month lead, about $1,600/kW EPC turnkey (Net Power 8-K (Net Power Inc. 2026), Observed, grade B), Modelled delivered cost about $58/MWh at 85% capacity factor and $4/MMBtu gas. A 24-engine, 429 MW order for a US data-center plant is the scale reference.

Aeroderivative turbines (LM2500/LM6000 class): 30–36 month lead, about $1,800/kW equipment, Modelled delivered cost near $200/MWh simple-cycle. Crusoe ordered 29 LM2500XPRESS units (near 1 GW) for Abilene.

Solid-oxide fuel cells (Bloom Energy (Company and press reporting 2026)): about 90-day deployment from manufacturing release, 1.5 GW cumulative deployed, $3,000–4,000/kW, Modelled delivered cost about $105/MWh. Viable where a gas lateral exists; capex is roughly double recip engines.

Rental generation: days to weeks, but $320–400/MWh at continuous duty (Modelled, grade C): a bridge, not a solution. At full 500 MW duty over 11.5 months the present value is $1.06B against a 12-month wait of $858M.

Not viable inside three years: new combined cycle (turbine slots reserved into 2031; GE Vernova's backlog is 116 GW with 2031 delivery reservations), large nuclear, SMRs (none operating in the West; first-of- a-kind cost about $18,700/kW at the Darlington final investment decision

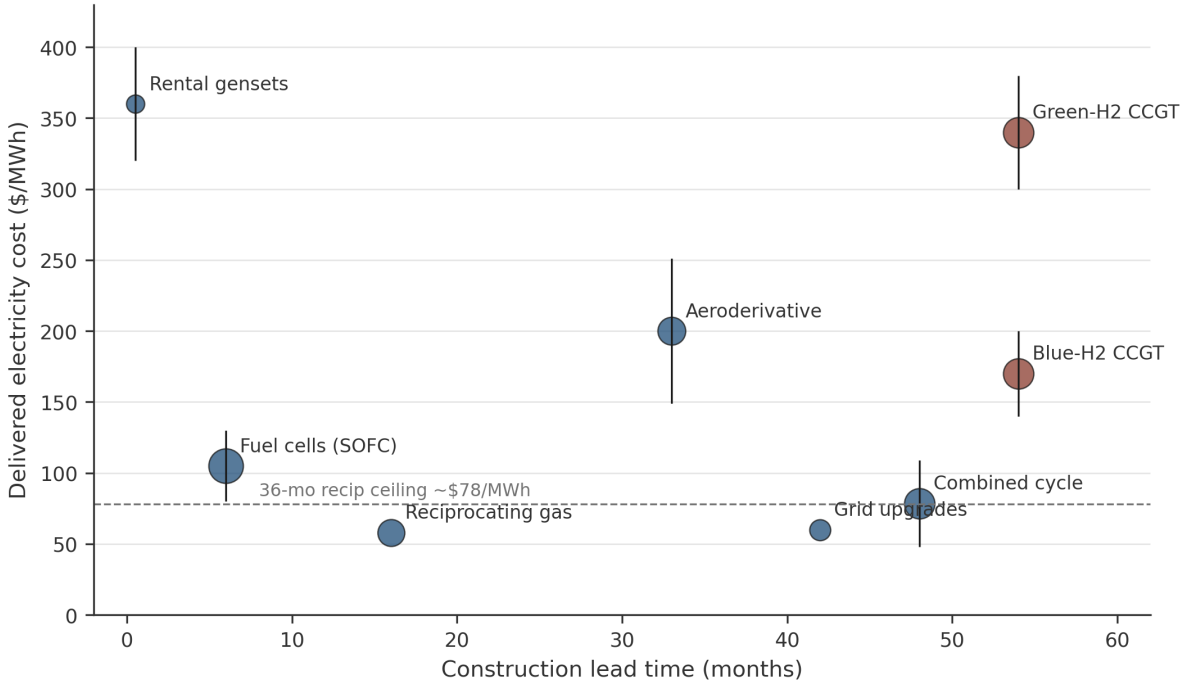


Figure 3: Alternatives by delivered electricity cost against construction lead time. Circle area scales with capex per kW. Only alternatives priced near grid power can clear the market for speed; hydrogen chains are a decarbonization option, not a time-to-power option.

(Energy industry press 2026)), enhanced geothermal at 500 MW scale, and any hydrogen chain. Solar-plus-storage can reach 500 MW inside three years but cannot carry a baseload profile without multi-day storage the headline LCOE excludes.

For a 2 GW need over ten years the ordering changes: customer-paid grid upgrades where headroom exists ($100–800/kW observed benchmarks, Observed, grade B), then gas combined cycle plus a pipeline lateral ($48–109/MWh Lazard (Lazard 2025), $2,400+/kW at 2026 backlog pricing, plus about $2.6/MWh for a representative 20-mile 24-inch lateral at 85% capacity factor), with solar/wind-plus-storage as an energy hedge ($44–131/MWh headline, but 2–3× oversizing for firm load).

Hydrogen, full chain. Built stage by stage, green hydrogen (electrolysis at $1,900–2,550/kW installed, 48–55 kWh/kg, $35/MWh input power (U.S. Department of Energy 2025)) reaches $4.67/kg plant-gate (Modelled), about $5.5–7.0/kg delivered after compression, storage, and 200 km pipeline transport, which is $300–380/MWh of electricity at 55% CCGT efficiency and $412–525/MWh at 40% simple-cycle efficiency. Blue hydrogen (SMR/ATR with carbon capture, $1.8–2.2/kg plant-gate) delivers electricity at about $140–200/MWh. Neither chain beats a $48–109/MWh gas combined cycle on cost. A 30%-by-volume hydrogen blend is only about 12% by energy; trucked hydrogen adds £2.2–3.9/kg under 300 km. Hydrogen is a decarbonization option, not a cheap-power option, and not a time-to-power option at these lead times.

5.5 Wait vs solve: the break-even boundary

Table 4 reports the maximum rational capex premium per kW of accelerated capacity. At 3, 6, and 12 months, nothing beats waiting for full replacement: every permanent alternative either arrives after the wait ends or carries a lifetime cost above the delay it cures.

Reciprocating gas engines are the only alternative that beats waiting inside 60 months, from a break-even horizon of 24.0 months by bisection. That figure is flattered by the stylized 24-month debt maturity: the m4 refinancing cliff adds about $300M to $W(24)$ exactly at the

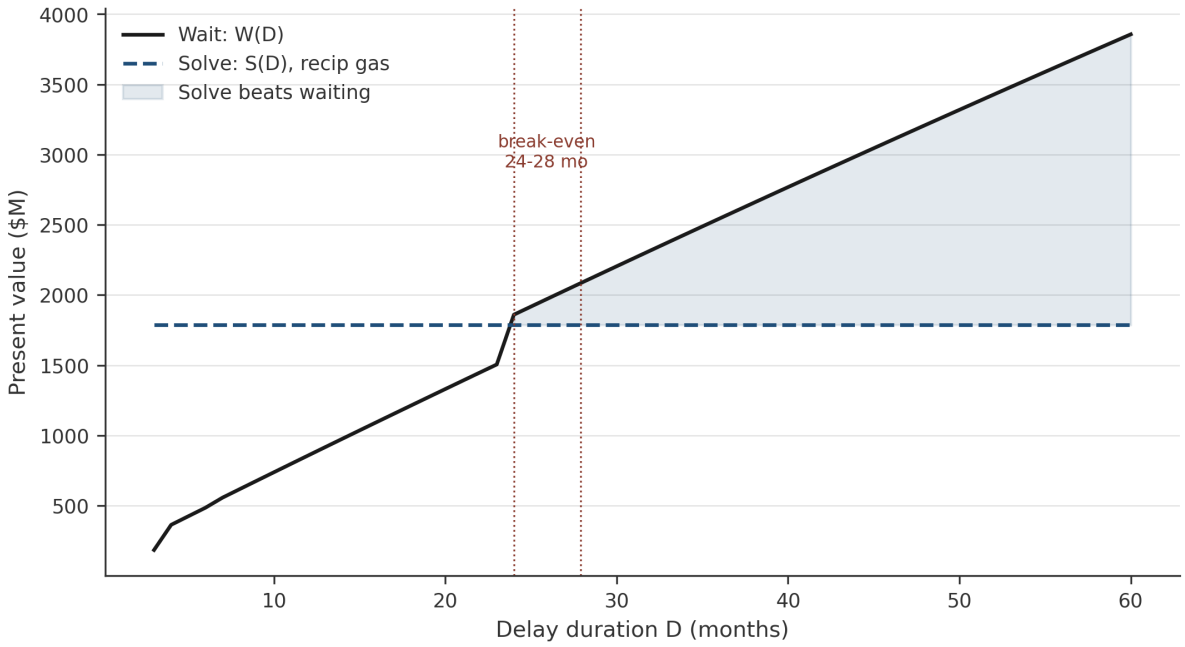


Figure 4: Present-value cost of waiting $W(D)$ against the cost of funding reciprocating gas engines directly, $S(D)$, for the stylized 500 MW / $6.0B data center. The crossing is at roughly 24–28 months; below it, waiting wins. Shaded bands show sensitivity to revenue intensity.

Table 4: Maximum rational capex premium ($/kW) for time-to-power, stylized case.

Alternative	3 mo	6 mo	12 mo	24 mo	36 mo
Recip gas	−1,672	−1,070	−325	1,680	3,049
Aeroderivative	−14,019	−13,417	−12,672	−10,668	−9,298
Fuel cells	−4,173	−3,571	−2,826	−821	548
Grid upgrades	−5,391	−4,789	−4,044	−2,039	−670
Gas combined cycle	−7,228	−6,626	−5,881	−3,876	−2,507

Notes: Negative means no premium is rational. Modelled, grade C.
Source: Author's model, `results/wait_vs_solve.py`.

crossing. With the cliff disabled, D^* moves to 27.9 months. The honest statement is that recip engines break even at roughly 24–28 months of expected delay. At 36 months the rational premium is $3,049/kW against $1,600/kW market capex: the project could pay nearly double the market price of the engines and still beat waiting. The maximum sustainable energy premium is +$18.1/MWh over grid, a delivered ceiling near $78/MWh. Fuel cells never beat waiting inside 60 months despite a $2.06B delay saving at 36 months, because their $105/MWh energy premium over 20 years ($1.79B present value) plus $1.72B capex loses by $1.45B. Grid upgrades break even only past 45.7 months, irrelevant to any wait under four years given their 42-month lead.

The structural result: for permanent assets, the 20-year fuel term dominates the delay saving. Capex differences between alternatives are second-order next to delivered $/MWh. Only electrons priced near grid clear the market for speed.

5.6 Real cases through the model

xAI Colossus. Modeled 12-month waiting cost is $258M at stylized revenue intensity, $460M at 3×, $662M at 5×; the rental bridge premium for 150 MW is $27.9M/month, $321M over 11.5

months. At stylized intensity the bridge costs more than waiting; at 3× intensity it costs less. The observed self-build is consistent with the model only if xAI's compute revenue per MW is a multiple of stylized lease revenue. xAI's mobile gensets were contracted or owned, likely below 24/7 rental pricing, so the rental figure is an upper bound.

Project Jupiter, New Mexico. Under stated assumptions ($25.7B capex, 70% debt, $4.9B/yr revenue at stylized intensity, 50% margin, all grade C), Modelled $W(6)$ is $2.18B, or $149k/MW/month, close to the stylized $143k. This is conditional on the pipeline slip becoming an energization slip, which is not established. An alternate 17.8-mile pipeline route at the FERC benchmark of $12.1M/mile (Observed, grade B) (Federal Energy Regulatory Commission 2025b) would cost about $215M (Modelled, grade C) against the conditional $2.18B waiting cost; its lead time is unknown because permitting is the constraint. Gas-fired alternatives do not solve a gas-delivery bottleneck, which narrows the solve set to waiting, rerouting, or non-gas bridge power.

Agratas Somerset. Under stated assumptions (55% debt, $2.5B/yr revenue, 25% EBITDA margin, all grade C), Modelled $W(12)$ is $887M at full exposure and $712M with temporary connections covering 40% of load. The mitigation cuts only 20% of the cost because interest, inflation, and idle-equity terms do not scale with temporary capacity: partial grid access is a weak substitute for the substation.

5.7 Sensitivity

Doubling revenue intensity raises $W(12)$ by 39% (to $1.19B); moving debt share from 50% to 85% moves $W(12)$ by less than 1%. Revenue intensity dominates; capital structure is second-order at these parameters, because higher debt raises interest terms while lowering the idle-equity term and the discount rate in offsetting directions. The $/MWh ceiling moves about 1:1 with the $60/MWh grid comparator (Assumed); the capex break-even moves through the energy premium.

Revenue and fuel-price symmetry. Table 5 recomputes the wait-vs-solve boundary under symmetric revenue and fuel-price moves, holding every other parameter at the stylized project. The break-even horizon is robust to revenue: halving revenue moves D^* from 24.0 to 24.7 months; doubling revenue leaves it at 24.0, with $W(12)$ moving from $689M to $1.19B around the $858M base. (Waiting cost does not depend on the gas price: W is fuel-independent.) The horizon is not robust to fuel price: at $6/MMBtu gas the recip-engine delivered cost rises to $72/MWh, D^* moves to 32.2 months, and the 36-month rational premium falls from $3,050/kW to $1,960/kW, still above the $1,600/kW market EPC price. Under the joint downside (half revenue, $6 gas), D^* reaches 36.9 months and the 36-month premium falls to $1,460/kW, below market EPC, so no rational premium exists; the $72/MWh engine no longer fits under the $71/MWh delivered-energy ceiling. The thesis survives each downside alone but fails under the joint downside. One softening at the other end: at $3/MMBtu gas the 12-month premium turns marginally positive ($226/kW, about 14% of market EPC), so the "nothing beats waiting at 12 months" result holds at $4 gas and above but not at $3.

6 Thesis determination

The brief asked for the numbers and for a verdict: is there a durable business thesis in monetizing time-to-power, who would capture it, and what would prove it wrong? This section works through it.

6.1 Revenue intensity decides who pays

The model is decisive on one point: revenue intensity dominates every other parameter. Doubling revenue intensity raises $W(12)$ by 39%; debt share from 50% to 85% moves it by less than

Table 5: Wait-vs-solve boundary under revenue and fuel-price scenarios.

Scenario	$W(12)$	$W(36)$	D^* (mo)	36-mo premium
Base: \$1.0B revenue, \$4 gas	\$858M	\$2.54B	24.0	\$3,050/kW
0.5× revenue, \$4 gas	\$689M	\$2.07B	24.7	\$2,540/kW
\$6 gas, \$1.0B revenue	\$858M	\$2.54B	32.2	\$1,960/kW
Joint: 0.5× revenue, \$6 gas	\$689M	\$2.07B	36.9	\$1,460/kW
2× revenue, \$4 gas	\$1.19B	\$3.49B	24.0	\$4,060/kW
\$3 gas, \$1.0B revenue	\$858M	\$2.54B	23.1	\$3,600/kW

Notes: $W(D)$ is the present-value waiting cost; D^* the recip-engine break-even horizon; the premium is the maximum rational 36-month capex premium against \$1,600/kW market EPC. Recomputed 2026-09-29; base case reproduces the paper's \$857.6M / 24.0 mo / \$3,049/kW.
Source: Author's model, `results/downside_sensitivity.py` and `results/downside_sensitivity.csv`.

1%. A data center at \$2.0M/MW/yr burns about \$143k per MW per month of delay; an aluminum smelter or green steel plant at one-fifth that intensity burns a fifth of it. Time-to-power is not a uniform commodity. It is worth ten times more to a GPU cluster than to a tonnage commodity plant, which is why xAI built its own power while industrial users queue.

The xAI case pins this down: a full-price rental bridge beats waiting only when compute revenue intensity is 2–3× the stylized lease assumption. Below that, the owner should wait; above it, the owner should buy power. Revenue intensity is the switch.

6.2 The stranded-capital pool is real but narrow

Three project-level costs are attributable to energy delays within the database's auditable rows: Alcoa's guided \$90–110M net loss in 2025, First Quantum's \$25M of additional 2024 power cost (Observed), and a \$50M conditional Project Jupiter term (6 months × \$143k/MW/mo × 390 MW of energization at risk, 50% probability weight, Modelled, grade C). The attributable pool is \$165–185M across three cases. Every other energy case lacks an auditable money loss.

Annualizing covered cases at their modeled run rates (not recurring totals): the data-center run rate is \$5.10B/year including the conditional Jupiter term, and \$0.74B/year excluding it. For scale, an illustrative Grade C scenario of 5 GW delayed for 12 months at \$143k/MW/month gives \$8.6B/year. The demand-driven queue evidence (55 months median, 2,061 GW in queue, 19% build rate) says the pool is large; the auditable money-loss evidence says it is thinly documented. Both are stated.

6.3 The bottleneck hierarchy

The database and the bottleneck evidence order the constraints by how much they gate energization. The utility grid connection is the primary gate (queue position, transformer slot, substation build), fuel supply is the secondary gate where the constraint is molecules rather than wires (Alcoa's power prices, First Quantum's drought-curtailment), and equipment availability (transformers at 48–60 month Tier-1 quotes, turbines at 5–7 year quotes) is the tertiary gate that lengthens every workaround. Five links run 30 months or more; three run 55 months or more. No credible monetization of time-to-power can assume any of them away.

6.4 Willingness to pay and who captures it

The order-of-magnitude willingness to pay for speed is \$143/kW per month of acceleration (the stylized \$143k/MW/month; xAI's observed bridge premium is \$186/kW/month at 150 MW scale). At 36 months of expected delay that supports a rational capex premium of \$3,049/kW and a delivered-energy ceiling near \$78/MWh.

Who captures it depends on the gate. Where the gate is the utility queue, the capture flows to whoever holds queue position or can build behind the meter: engine OEMs and packagers (Wärtsilä's 429 MW order, GE Vernova's 29-turbine Crusoe order), fuel-cell deployers with 90-day lead times (Bloom), and the IPPs that assemble power-park products. Where the gate is molecules, it flows to pipeline rerouting and firm gas. The durable moat is not any single technology; it is a repeatable capability to deliver grid-cost-adjacent electrons in 14–30 months, which today only reciprocating gas plants credibly offer at data-center scale.

6.5 The verdict

The evidence supports a durable but narrow thesis: there is a monetizable business in time-to-power, but it is confined to three conditions. (1) Delays beyond roughly two years, where recip engines beat waiting at 24–28 months and grid upgrades only clear at 45.7 months. (2) Alternatives priced near grid power: only electrons near $60–78/MWh beat waiting, because the 20-year fuel term dominates the delay saving. (3) Revenue-intense projects: the willingness to pay collapses with revenue intensity, and the capital-structure effect is second-order.

What the evidence does not support: a general thesis that energy-delayed megaprojects are broadly worth paying to accelerate; a hydrogen business case for time-to-power (even the cheaper blue chain costs $140–200/MWh against a $78/MWh ceiling); and any value for SMR or new nuclear as speed products (72–120 month lead times put them outside every modeled horizon).

6.6 Falsifiers: what would prove this wrong

These are the dashboard readings that would invalidate the thesis as stated, with thresholds where the evidence supports them:

1. The median 200+ MW generator request-to-operation time falls below 36 months; below 24 months breaks the grid-connected thesis outright.
2. Large transformer lead times normalize to 18 months or less, removing the equipment gate.
3. Aggregate hyperscaler capex guidance falls at least 20% year over year from the $357.5B 2025 base, breaking the demand premise.
4. US data-center vacancy exceeds 5%, or lease-rate growth is negative for two consecutive quarters (against 1.4% vacancy and +6.5% growth last observed in the companion buildout study).
5. New 100+ MW loads demonstrably energize inside 24 months after FERC reform, showing the queue gate has lifted.
6. Heavy-frame turbine order-to-COD quotes fall below 36 months, making combined cycle a speed product.
7. Delayed-project debt returns to par without schedule improvement (the 89–91 cent Project Jupiter quote is the reference), showing the market has stopped pricing delay risk.
8. Continuous-duty rental generation falls below $150/MWh, making bridges cheap enough to fund at scale.

7 Robustness and limitations

Stylized, not appraised. The stylized project's revenue, margin, capex, and capital structure are stated assumptions, not observations of any real asset. The $2.0M/MW/yr revenue assumption is the most consequential Grade C input: it uses the low end of CBRE asking rents, not transacted rents, and moving it moves every break-even.

The 24-month crossing. The modeled 24.0-month break-even for recip engines is flattered by the 24-month debt maturity: the m4 refinancing cliff fires exactly at the crossing. With the cliff disabled the crossing is 27.9 months. We report the range, not the point.

Grade C scaffolding. The delay probabilities in the bottleneck dataset, the grid comparator price ($60/MWh), the equity required return (10%), the margin step-up (250 bps), the waiver and refinancing parameters, and the EBITDA margin (50%) are Assumed. The model is explicit about each one; none is presented as measurement.

What is not modeled. Tax shields (interest deductibility would reduce after-tax waiting cost); retiring the alternative once grid power arrives (would lower solve cost, conservative against the thesis); correlation between inflation, rates, and repricing in stress; the m7/m8 split for fixed-end-date leases; carbon pricing; and site-specific permitting for the alternatives.

Evidence limits. Most project rows are Observed (grade B), not primary. Project Jupiter has no located SEC filing, county bond record, or primary permit document; its campus target has not slipped and the $2.18B modeled cost is conditional. Agratas causality between substation and production timing is not established. The mining and Alcoa rows lack backup-capex figures. Transformer and turbine pricing are market snapshots, not long-run costs.

What would change the answer. The single most valuable datum not publicly available is transacted data-center lease revenue per MW under energization-linked contracts; it would replace the most consequential assumption in the model.

8 Conclusion

When a financed asset cannot start because the energy is not there, the cost of waiting is mostly not interest. For the stylized data center, a 12-month delay costs $857.6M in present value, and only 35% of that is financing terms; the rest is the timing value of revenue, inflation on unspent capex, and the cost of idle equity. Counting interest alone understates the cost by roughly two-thirds.

Against that cost, the alternatives sort cleanly. Nothing beats waiting at 12 months or less. Past roughly two years, reciprocating gas engines beat waiting, and they beat it decisively enough to justify paying nearly double their market price. Everything else either arrives too late (grid upgrades, combined cycle, nuclear, geothermal, hydrogen) or costs too much per electron over its life (fuel cells, rentals, aeroderivatives) to justify full replacement inside any modeled horizon.

The thesis is therefore conditional, not general: time-to-power is worth paying for where delays run beyond two years, where the alternative's electrons price near grid, and where the project's revenue intensity justifies the premium. Those conditions currently describe one asset class well, the AI data center, and little else. The falsifiers in Section 6.6 say exactly what would change that.

9 Dataset and reproducibility

All datasets and models are published alongside this paper (all paths relative to the study root):

- `data/master_projects.csv`
- `data/master_energy_bottlenecks.csv`

- `data/master_delay_mechanics.csv`
- `data/master_alternatives.csv`
- `results/delay_cost_model.py`
- `results/wait_vs_solve.py`
- `results/key_estimates.csv` (103 estimates)

Production note. This paper was typeset in LaTeX (11pt, 1-inch margins) from `paper/paper.tex`; all figures are 300 DPI PNGs under the shared institutional chart style; the delay-cost and wait-vs-solve engines are the seeded `results/delay_cost_model.py` and `results/wait_vs_solve.py`. The PDF is the version of record. Figures are generated by `paper/make_figures.py` from the audited datasets. The claim index is `paper/CLAIM_INDEX.md`; the audit script is `paper/audit_claims.py`. All monetary values are nominal USD, base year 2026, unless stated. Research design, statistical analysis, and drafting were carried out through a quantitative research pipeline with AI assistance; every estimate, figure, and quantitative claim was verified against the source data before publication.

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